



TECHNISCHE
UNIVERSITÄT
DARMSTADT

Information About New Device in Your Office

Dear Office-Neighbour,

I'm reaching out to introduce you to a new device recently installed in your office as part of our ongoing research. Since it's in your immediate vicinity, I'd like to share the details transparently.

The device is a node of the **Shepherd Nova** Testbed. As part of cfaed- and REC2-cluster, our NES Lab research group is building the Testbed. Shepherd is a public developing platform for rigorous experiments under repeatable energy-harvesting conditions. The testbed helps explore ways to run smart devices without the need for batteries and their potentially harmful chemistry.

Your office was selected due to its location and its contribution to the overall RF network we're setting up.

What is the device itself? It's called an "observer node" and here are its key characteristics:

- **Connection & Power:** It connects and draws power via Ethernet. Its power consumption is very low, typically less than 5 W, so it won't noticeably impact your office's environment
- **Privacy:** There are no embedded sensors or microphones – so users of the testbed cannot snoop on the local staff. It's purely a communication and networking component.
- **RF:** The embedded target board contains an RF frontend with an antenna, operating on the 2.4 GHz ISM band, which is also used by Bluetooth, older Wi-Fi devices and common kitchen microwaves. A device operating in the same spectrum, like wireless headphones, could in theory be interfered with, but in our case, it is highly unlikely. In our experiments we simulate energy-scarce environments, so the firmware will mostly sleep and listen (receive), as transmissions are relatively energy-intensive. In addition, transmit power is usually reduced to save energy and is limited to a maximum of 10 mW, which is less than or equal to that of a common phone.

As the project is completely open source, more information is available online at testbed.nes-lab.org.

If anything seems unusual, or if you have any questions, suggestions, or concerns at all, please don't hesitate to contact me. I'm happy to chat.

You can reach me at ingmar.splitt@tu-dresden.de

Thanks for your time and understanding!

Best regards,
Ingmar Splitt

Fachbereich
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Datum
27.03.2026



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